Experiment No. 24 – Black Hat Morphological Operation

1. Aim

To implement the Black Hat technique as a Morphological Operation using OpenCV to extract small dark details from the original image.

2. Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()
filename = list(uploaded.keys())[0]

# Read image
image = cv2.imdecode(
    np.frombuffer(uploaded[filename], np.uint8),
    cv2.IMREAD_COLOR
)

if image is None:
    print("Image loading failed")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply Top Hat operation
tophat = cv2.morphologyEx(
    gray, cv2.MORPH_TOPHAT, kernel
)

# Display output
plt.figure(figsize=(12, 5))

plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(tophat, cmap="gray")
plt.title("Top Hat Image")
plt.axis("off")
```

```
plt.tight_layout()
plt.show()
```

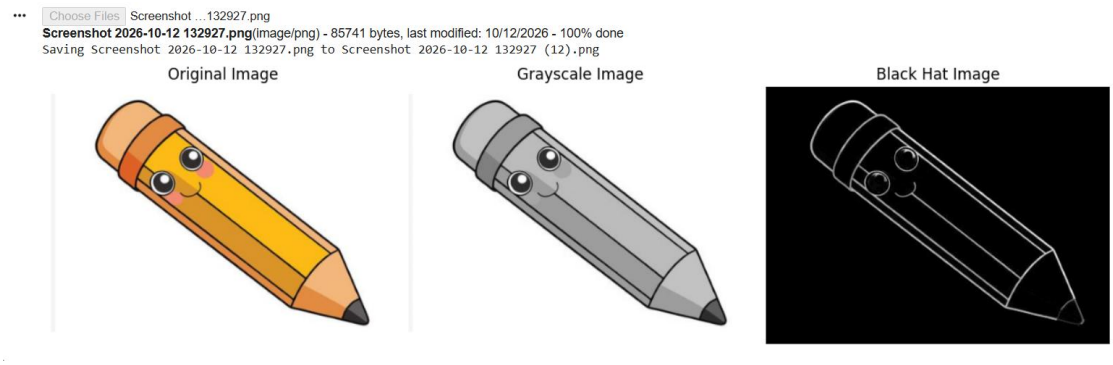
3. Expected Output

The output displays:

- Original Image.
- Grayscale Image.
- Black Hat Image.

The Black Hat operation highlights small dark regions and details in the image.

4. Output



The Black Hat morphological operation is successfully applied to extract dark details from the image.

5. Result

Thus, the Black Hat Morphological Operation was successfully implemented using Python and OpenCV.